1. **Problem**
   Numeric answer, give your answer with a precision of 2 decimal place(s).

   This is the first question in a set of questions concerning the mass of offspring.

   Introduction: You are given a dataset about the mass of offspring, the sex of offspring, and the mass of the mother. There are three variables in the dataset: `Offspring_mass`, `Offspring_sex`, and `Mother_mass`.

   Import the dataset `offspring_mass_2023.csv`. What is the mean mass of female offspring?

   If you need RStudio server to answer the question, click **here**.

   Solution
   10.08

2. **Problem**
   Numeric answer, give your answer with a precision of 2 decimal place(s).

   What is the value of the total sum of squares of the offspring mass variable?

   If you need RStudio server to answer the question, click **here**.

   Solution
   106.87

3. **Problem**
   Multiple choice.

   Make an appropriate visualisation of the data (offspring mass is the response variable). Inspect the visualistion and then answer which of the follow you expect to be true. Be careful, some answer might not be as simple as they seem.

   If you need RStudio server to answer the question, click **here**.

   (a) The intercept for the male regression line is around 10.8
   (b) Female offspring mass tends to be greater than male offspring mass.
   (c) There is weak or no evidence of an interaction between mother mass and offspring sex.
   (d) Offspring mass appears to be positively related to mother mass.

   Solution

   - Offspring mass appears to be positively related to mother mass.
   - There is weak or no evidence of an interaction between mother mass and offspring sex.

4. **Problem**
   Numeric answer, give your answer with a precision of 0 decimal place(s).

   Fit to the data a linear model with normal errors, both explanatory variables, and the interaction between the two explanatory variables.

   How many degrees of freedom for error are there?

   If you need RStudio server to answer the question, click **here**.

   Solution
   63

5. **Problem**
   Numeric answer, give your answer with a precision of 4 decimal place(s), and give the absolute value.

   What is the estimate difference in the slope of male and female offspring?

   If you need RStudio server to answer the question, click **here**.

   Solution
   0.0054

6. **Problem**
Single choice.

In this study of offspring mass, assume that we collected and measured the mothers and their offspring in the wild. In the data we see a strong relationship between mother mass and offspring mass. Can we therefore conclude that variation in mothers size is the cause of the variation in offspring size?

If you need RStudio server to answer the question, click **here**.

(a) Yes

(b) No

Solution
No

7. **Problem**
Single choice.

Imagine that we had recorded the number of offspring produced by each mother as the response variable. What type of model would be most appropriate?

If you need RStudio server to answer the question, click **here**.

(a) A mixed effects model, mother as the random effect.

(b) Segmented regression, with one segment for each mother.

(c) ANCOVA, with mother as the categorical explanatory variable.

(d) A linear model with Poisson error family.

Solution
A linear model with Poisson error family.

8. **Problem**
Numeric answer, give your answer with a precision of 2 decimal place(s).

Imagine that we had found that our response variable has a total sum of squares equal to 43.05 and that the sum of squares left unexplained by our model was 29.45. What would the coefficient of determination of our model be?

If you need RStudio server to answer the question, click **here**.

Solution
0.32

9. **Problem**
Numeric answer, give your answer with a precision of 1 decimal place(s).

In a multiple regression model, a biologist wants to assess collinearity among explanatory variables using variance inflation factors (VIFs). For one explanatory variable, a regression of that explanatory variable on all the other explanatory variables gives an R^2 value of 0.75.

What is the VIF for this explanatory variable?

If you need RStudio server to answer the question, click **here**.

Solution
4.0

10. **Problem**
Single choice.

Which of the assumptions of a linear model is usually checked by looking at the QQ-plot?

If you need RStudio server to answer the question, click **here**.

(a) Normally distributed residuals.

(b) Homoscedasticity.

(c) Independence.

(d) Linearity.

Solution
Normally distributed residuals.

11. **Problem**
Single choice.

If we only look at the F-statistic (and its associated p-value) of the full model (which in this case is 121.39), what can we determine to be true?

If you need RStudio server to answer the question, click **here**.

(a) The model explained a significant amount of variation in offspring mass.

(b) There was a strong association between offspring sex and offspring mass.

(c) There was a strong association between mother mass and offspring mass.

(d) There was a strong association between offspring sex and mother mass.

Solution

- The model explained a significant amount of variation in offspring mass.

12. **Problem**
Multiple choice.

When communicating the results of the study, which of these pieces of information should be included?

If you need RStudio server to answer the question, click **here**.

(a) The value of the test statistic.

(b) A statement about any outliers that were removed from dataset.

(c) The amount of variation explained by the model (i.e. the r-squared).

(d) The number of degrees of freedom for error.

Solution
All should be included

13. **Problem**
Single choice.

Imagine that the offspring in this study came from 10 mothers. In what way should we modify our analysis?

If you need RStudio server to answer the question, click **here**.

(a) Use a mixed model.

(b) Use non-linear regression.

(c) Use time series analysis.

(d) Use path analysis.

(e) Switch to a Bayesian analysis.

Solution

- Use a mixed model.

14. **Problem**
Single choice.

This is the first question in a set of questions concerning a case study about the effect of two drugs.

Introduction to case study: Imagine you are approached by a colleague that is researching the effects of two drugs: drug A and drug B. They made an experiment with four treatment combinations: control (no drug A and no drug B), only drug A, only drug B, and both drug A and B. They applied these treatments to cultures of human cells and observed if the culture survived or died. This resulted in a dataset with four variables: `Drug_A`, `Drug_B`, `Replicate`, and `Outcome`. The two drug variables contain the value `absent` or `present` depending if the drug was not applied (absent) or was applied (present). The `Replicate` variable indicates which replicate a row of the dataset concerns. The `Outcome` variable is binary (0/1) with 0 indicating the culture died, and 1 indicating that it survived.

Import the dataset `drug_effect_2023.csv`.

In which treatment combination did the greatest proportion of replicates have cell which survived?

If you need RStudio server to answer the question, click **here**.

(a) Drug A present and Drug B present

(b) Drug A absent and Drug B present

(c) Drug A absent and Drug B absent

(d) Drug A present and Drug B absent

Solution

- Drug A present and Drug B absent

15. **Problem**
Multiple choice.

You will need to use a binomial generalised linear model to analyse the data. Why is this type of model required? Be sure to indicate all of the correct answers and recall that none, one, or more than one may be correct.

If you need RStudio server to answer the question, click **here**.

(a) The Drug A variable is binary (i.e. either absent or present).

(b) The Outcome variable is binary (i.e. either alive or dead).

(c) The Drug B variable is binary (i.e. either absent or present).

Solution

- The Outcome variable is binary (i.e. either alive or dead).

16. **Problem**
Multiple choice.

Using a graph, decide which of the following are correct.

If you need RStudio server to answer the question, click **here**.

(a) When drug A is absent then drug B appears to have a positive effect on survival.

(b) When drug A is present then drug B appears to have a positive effect on survival.

(c) There appears to be some evidence of an interaction effect of the two drugs.

(d) When drug B is absent then drug A appears to have a positive effect on survival.

Solution

- When drug B is absent then drug A appears to have a positive effect on survival.
- When drug A is absent then drug B appears to have a positive effect on survival.
- There appears to be some evidence of an interaction effect of the two drugs.

17. **Problem**
Numeric answer, give your answer with a precision of 5 decimal place(s).

Consider only the part of the dataset for which Drug B is absent. What is the value of the odds ratio that compares the drug A absent and present treatment?

Hint: If the presence of drug A increases survival, we would expect the odds ratio to be greater than one.

If you need RStudio server to answer the question, click **here**.

Solution
60

18. **Problem**
Numeric answer, give your answer with a precision of 0 decimal place(s).

Fit a model to the data, with the response variable giving the outcome, and the two explanatory variables (Drug_A and Drug_B) and the interaction between them. You should see that there are 68 degrees of freedom for error. How many degrees of freedom for error would there be if another 23 observations were added, and another drug treatment was added, i.e., Drug_C with levels *absent* and *present*. Assume that the experiment and therefore model includes all possible treatment combinations.

If you need RStudio server to answer the question, click **here**.

Solution
87

19. **Problem**
Numeric answer, give your answer with a precision of 2 decimal place(s), and give the absolute value.

For the previous question you should have fit a model to the data, with the response variable the outcome of alive or dead, and the two explanatory variables (Drug_A and Drug_B) and the interaction between them. What is the estimated difference, on the scale of the response variable (here a probability), between the treatment combination Drug_A absent + Drug_B absent and the treatment combination Drug_A present + Drug_B present.

If you need RStudio server to answer the question, click **here**.

Solution
0.03

20. **Problem**
Numeric answer, give your answer with a precision of 2 decimal place(s).

In the model you fit to the data (including the interaction term) what amount of deviance is explained by the Drug_A term?

If you need RStudio server to answer the question, click **here**.

Solution
1.01

21. **Problem**
Multiple choice.

Which of these statements is true?

If you need RStudio server to answer the question, click **here**.

(a) The amount of over-dispersion can be given by dividing the deviance explained by the model and the residual degrees of freedom.

(b) When over-dispersion is present, it can be reduced by including new and important explanatory variables in the model.

(c) In this case study there is no reason to be concerned about over-dispersion.

(d) Overdispersion can be corrected by altering the error distribution family used in the model.

Solution

- In this case study there is no reason to be concerned about over-dispersion.
- When over-dispersion is present, it can be reduced by including new and important explanatory variables in the model.
- Overdispersion can be corrected by altering the error distribution family used in the model.

22. **Problem**
Numeric answer, give your answer with a precision of 2 decimal place(s).

The binomial model described in question 20 involves a linear predictor. If an estimated value of the linear predictor is -1.2, what would be the corresponding probability of a culture surviving?

Hint: Assume that the default link function is being used.

If you need RStudio server to answer the question, click **here**.

Solution
0.23

The default link function for a binomial GLM is the logit link, so the linear predictor is on the log-odds scale. To convert a linear predictor back to a probability, use the inverse-logit transformation:

$$p = \frac{\exp(-1.2)}{1 + \exp(-1.2)} = 0.23.$$

The corresponding probability that a culture survives is therefore 0.23.

23. **Problem**
Multiple choice.

In the analyses we so far made, we assumed that each replicate could be considered to be an independent data point. Why might this assumption be unwise?

If you need RStudio server to answer the question, click **here**.

(a) If all the replicates of a treatment combination were placed in their own incubator.

(b) If each replicate was not derived from a separate and independent mother cell line.

(c) Because there were different numbers of replicates in different treatment combinations.

(d) If some replicates were done on one day, and others on another day.

Solution

- If some replicates were done on one day, and others on another day.
- If all the replicates of a treatment combination were placed in their own incubator.
- If each replicate was not derived from a separate and independent mother cell line.

24. **Problem**
Multiple choice.

Imagine that instead of looking if the replicates were dead or alive, we had measured the time until death occurred. Which of the following would be true?

If you need RStudio server to answer the question, click **here**.

(a) The response variable is no longer binary, but is rather a count.

(b) We should still use the logit link function.

(c) We should use survival analysis on these type of data.

(d) Such data could be used to estimate average time to death in each treatment combination.

Solution

- We should use survival analysis on these type of data.
- Such data could be used to estimate average time to death in each treatment combination.

25. **Problem**
Multiple choice.

Imagine that we had measured more variables and were going to use them in a mixed model. Which of the following variables would we include as a **random effect**?

If you need RStudio server to answer the question, click **here**.

(a) The temperature of the replicates during the experiment.

(b) The identity of the one person that performed the experiment.

(c) Which of ten cell line that each replicate was derived from.

(d) The position of each replicate in the multi-well plates used in the experiment.

Solution

- Which of ten cell line that each replicate was derived from.
- The position of each replicate in the multi-well plates used in the experiment.

26. **Problem**
Single choice.

Imagine that numerous other research groups have performed the same experiment. What type of analysis would be most appropriate to analyse the results of all the experiments?

If you need RStudio server to answer the question, click **here**.

(a) Meta-analysis.

(b) Bayesian analysis.

(c) Multivariate analysis.

(d) Time series analysis.

Solution

- Meta-analysis.

27. **Problem**
Numeric answer, give your answer with a precision of 2 decimal place(s).

What is the value of the t-statistic that tell us about whether the association between offspring and mother mass differs between female and male offspring?

If you need RStudio server to answer the question, click **here**.

Solution
0.12

28. **Problem**
Single choice.

What measure could we use to compare goodness of fit of a model with the interaction term and a model without the interaction term?

If you need RStudio server to answer the question, click **here**.

(a) Multiple r-squared.

(b) The difference in the estimated slopes.

(c) Adjusted r-squared.

(d) The number of degrees of freedom for error.

Solution

- Adjusted r-squared.

29. **Problem**
Multiple choice.

Imagine that we also measure other things about the effects of the drugs on the cells, for example, cell size, and the chemical composition of the cells. Which of the following statements is true.

If you need RStudio server to answer the question, click **here**.

(a) Controlling the experiment wide error rate (type-1 error rate) would be appropriate.

(b) A time series analysis would be appropriate.

(c) Multivariate analyses, so to reduce the number of tests performed, would be appropriate.

(d) Use of path analysis would be appropriate.

Solution

- Multivariate analyses, so to reduce the number of tests performed, would be appropriate.
- Controlling the experiment wide error rate (type-1 error rate) would be appropriate.
- Use of path analysis would be appropriate.

30. **Problem**
The remaining questions in the exam do not refer to a particular data set or case study. They cover material from across the course.

Multiple choice.

A biologist uses principal component analysis (PCA) to summarise variation in four measured environmental variables across many field sites. The first two principal components explain 71% of the total variation in the environmental variables. Site A and Site B have very similar scores on PC1 and PC2, while Site C is far from both of them in the PCA plot.

Which interpretations are appropriate?

(Multiple choice, 1 point)

If you need RStudio server to answer the question, click **here**.

(a) PCA is being used here to reduce several measured variables to a smaller number of summary axes.

(b) PC1 and PC2 together explain all variation in the original environmental variables.

(c) PCA proves that the environmental variables cause differences in species composition.

(d) Site A and Site B are probably similar in the main environmental gradients summarised by PC1 and PC2.

(e) Site C must have higher values for every original environmental variable than Sites A and B.

Solution

(a) True.

(b) False.

(c) False.

(d) True.

(e) False.

31. **Problem**
Multiple choice.

A researcher uses non-metric multidimensional scaling (NMDS) to visualise differences in community composition among sampling sites. The NMDS was fitted using a Bray-Curtis dissimilarity matrix calculated from species abundance data, and the final two-dimensional solution has stress = 0.09.

Which of the following statements are correct?

(Multiple choice, 1 point)

If you need RStudio server to answer the question, click **here**.

(a) Sites plotted close together in the NMDS ordination are represented as having more similar community composition than sites plotted far apart.

(b) The axes in an NMDS plot should usually be interpreted as the original measured species abundances.

(c) A stress value of 0.09 suggests that the two-dimensional ordination is a reasonably good representation of the dissimilarities.

(d) NMDS attempts to preserve the rank order of dissimilarities among sites, rather than modelling a response variable as a function of explanatory variables.

Solution

(a) True.

(b) False.

(c) True.

(d) True.

32. **Problem**
Multiple choice.

You have performed a regression analysis with *standardized* explanatory variables (i.e., mean zero and standard deviation 1) with data from an observational study. The regression coeffcients are $\hat{\beta}_x = 0.65$ (CI from -0.01 to 1.31) and $\hat{\beta}_z = 0.15$ (CI from 0.01 to 0.29). What are possible conclusions?

If you need RStudio server to answer the question, click **here**.

(a) The variable x is probably more relevant than z, because $|\beta_x| > |\beta_z|$.

(b) z is causally related to the response.

(c) The variable z is more relevant than x, because the p-value for β_z is < 0.05, but the p-value for β_x is > 0.05.

(d) Both coefficients have very similar p-values.

Solution

(a) True

(b) False.

(c) False.

(d) True.

33. **Problem**
Multiple choice.

Which of the statements are true?

If you need RStudio server to answer the question, click **here**.

(a) Researchers and statisticians are unanimous that the threshold significance of 0.05 is correct.

(b) Strictly speaking, we can never accept the null or the working hypothesis.

(c) A biological effect may exist, but we may find it is not statistically significant.

(d) A finding may be statistically significance but have little biological significance.

Solution

(a) False.

(b) True.

(c) True.

(d) True.

34. **Problem**
Single choice.

Assume that your null hypothesis (H_0) is $\beta_x = 0$, that is, the respective explanatory variable x has no effect on the outcome in a regression. Which of the following statements about the respective p-value in the regression output are true?

If you need RStudio server to answer the question, click **here**.

(a) (1-p) is the probability that $\beta_x > 0$.

(b) (1-p) is the probability that $\beta_x \neq 0$.

(c) The p-value is the probability to observe an estimate $\hat{\beta}_x$ that is at least as extreme as the one observed, given that $\beta_x = 0$.

(d) The p value is the probability that $\beta_x = 0$ (H_0 true).

Solution

(a) False.

(b) False.

(c) True.

(d) False.

35. **Problem**
Multiple choice.

Which of the following statements about Poisson regression are true?

If you need RStudio server to answer the question, click **here**.

(a) In Poisson regression the typical link function is the exponential function.

(b) If overdispersion is a problem, we can replace the Poisson model by a negative binomial model, because it no longer assumes that $E(Y) = Var(Y)$.

(c) Poisson regression is typically used when the response is a binomial variable (i.e., a count between 0 and a fixed number of trials).

(d) We need to look at the residual deviance and the degrees of freedom to check for over- or underdispersion.

Solution

(a) False

(b) True

(c) False

(d) True

36. **Problem**
Multiple choice.

In a study with 40 participants it was tested whether a novel treatment (`treat`) reduces the number of polyps (`number`) in the colon of the patients. The model included age and sex as additional predictors, and the output of the respective Poisson model is given as follows:

```
summary(polyps.glm)
```

```
##
## Call:
## glm(formula = number ~ treat + age + sex, family = "poisson",
##     data = polyps)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.001871   0.423961   0.004  0.99648
## treatdrug   -1.022359   0.329669  -3.101  0.00193 **
## age          0.022484   0.012864   1.748  0.08051 .
## sexmale      0.143101   0.267946   0.534  0.59330
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 48.632  on 39  degrees of freedom
## Residual deviance: 31.909  on 36  degrees of freedom
## AIC: 112.86
##
## Number of Fisher Scoring iterations: 5
```

Which of the following statements are correct?

If you need RStudio server to answer the question, click **here**.

(a) The treatment seems to be very effective.

(b) A 50 years old male that was not treated has an expected number of polyps of 3.56, while an untreated female of the same age has 3.08.

(c) The expected number of polyps increases by about 25% every ten years.

(d) The data is clearly overdispersed, thus we cannot trust the p-values.

Solution

(a) True

(b) True

(c) True

(d) False

37. **Problem**
Single choice.

For which would one use structural equation modelling?

If you need RStudio server to answer the question, click **here**.

(a) My study is about the length of time it takes for a lethal drug to take effect.

(b) I would like to analyse temporal patterns in cardiac rhythms.

(c) I would like to formally include some prior information I have about the answer to my question.

(d) Exploring and visualising variation and co-variation among multiple variables. Reducing the number of dimensions required to represent many variables.

(e) I would like to use an analysis that allows some of my variables to be both a response and an explanatory variable.

Solution

(a) False.

(b) False.

(c) False.

(d) False.

(e) True.

38. **Problem**
Single choice.

For which would one use a generalised additive model?

If you need RStudio server to answer the question, click **here**.

(a) My study is about the time it takes for a lethal drug to take effect.

(b) I have recorded information about the location and concentration of a dangerous pollutant.

(c) I want to model the association between two variables, but don't care about its mathematical form.

(d) I would like to calculate my statistical power of my planned study.

(e) I have been analysing a dataset for days, and seem to be getting nowhere.

Solution

(a) False.

(b) False.

(c) True.

(d) False.

(e) False.

39. **Problem**
Multiple choice.

Assume data is in the object `dd` the explanatory variable is `mass` and the response variable is `cholesterol_conc`. Which of these commands would make an appropriate model of this data?

If you need RStudio server to answer the question, click **here**.

(a) `lm(cholesterol_conc ~ mass, data=dd)`

(b) `lm(mass ~ cholesterol_conc , data=dd, family=poisson)`

(c) `lm(cholesterol_conc > mass, data=dd)`

(d) `lm(mass ~ cholesterol_conc, data==dd)`

Solution

(a) True.

(b) False.

(c) False.

(d) False.

40. **Problem**
Multiple choice.

Imagine you have a dataset containing the blood pressure of individuals treated with five different drugs. Which of these functions would you need to use to create a new variable that is the log of the blood pressures and then make a dataset that excludes individuals above a certain age?

If you need RStudio server to answer the question, click **here**.

(a) `filter`

(b) `arrange`

(c) `mutate`

(d) `group_by`

Solution

(a) True.

(b) False.

(c) True.

(d) False.